UNIVERSITY OF WAH(UOW)

DEPARTMENT OF COMPUTER SCIENCE

Lab Sessional

COURSE: Data Structure & Algorithm Lab_201L

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Task 1

```cpp
// 1)
int quantity;
float price;
float totalBill;

cout << "Enter Quantity: ";
cin >> quantity;
cout << "\nEnter Price: ";
cin >> price;
totalBill = quantity * price;
cout << "Total : " << totalBill << endl;
```

```cpp
// 2)
int a, b;
cout << "Sum: " << a + b << endl;
cout << "Sub: " << a - b << endl;
cout << "Multiply: " << a * b << endl;
cout << "Division: " << a / b << endl;
cout << "Modulo: " << a % b << endl;
```

```cpp
int marks;
cout << "Enter marks: ";
cin >> marks;
char grade;
if (marks > 90)
{
    grade = 'A';
}
else if ((marks > 80) && (marks <= 90))
{
    grade = 'B';
}
else if ((marks > 70) && (marks <= 80))
{
    grade = 'C';
}
else
{
    grade = 'F';
}
cout << "Your grade is: " << grade << endl;
```

```cpp
// 4)
for (int i = 1; i <= 10; i++)
{
    cout << "Number: " << i << endl;
}
```

```
// 5)
int n, sum = 0;
cout << "Enter Number: ";
cin >> n;
int i = 1;
while (i <= n)
{
    sum = sum + i;
    i = i + 1;
}
cout << "Sum is: " << sum << endl;
```

```
// 6)
int n, sum = 0;
cout << "Enter Number: ";
cin >> n;
int i = 1;
do
{
    cout << n << " * " << i << " = " << i * n << endl;
    i++;
} while (i <= 10);
```

Task # 02

```
int main(){
    int stock[10] = {5, 15, 22, 35, 40, 55, 65, 70, 85, 95};
    int n = 10;
    // ----------------------------------
    int s;
    cout << "Enter Number to Search: " << endl;
    cin >> s;
    for (int i = 0; i < 10; i++)
    {
        if (s == stock[i])
        {
            cout << "NuMber is founded." << endl;
            break;
        }
    }
```

```cpp
int p1;
cout << "Enter Position: " << endl;
cin >> p1;
for (int i = n; i > p1; i--)
{
    stock[i] = stock[i - 1];
}
stock[p1] = 66;
n++;
for (int i = 0; i < n; i++)
{
    cout << stock[i] << " ";
}
```

```cpp
int p1;
cout << "Enter Position: " << endl;
cin >> p1;
for (int i = 0; i < p1; i++)
{
    stock[i] = stock[i + 1];
}
n--;
for (int i = 0; i < n; i++)
{
    cout << stock[i] << " ";
}
// -------------------------------------------
// sizeof operator
int t = sizeof(stock);
cout << "Total size of operator is: " << t << endl;
```

```cpp
int max = stock[0], min = stock[0];
for (int i = 0; i < n; i++)
{
    if (max < stock[i])
    {
        max = stock[i];
    }
    if (min > stock[i])
    {
        min = stock[i];
    }
}
cout << "Max: " << max << endl;
cout << "Min: " << min << endl;
// ----------------------------------------------
```

Task # 03

```
struct student
{
    string name;
    int roll;
    int marks;
};

int main()
{
    // ---------------------------------Task # 03----------------------------
    int  students[3][3] = {
            {80, 85, 78},
            {88, 82, 90},
            {75, 89, 84}
        };
        // -----------------------------------------------
        for (int i = 0; i < 3; i++)
        {
            for (int j = 0; j < 3; j++)
            {
                cout<<students[i][j]<<" ";
            }
            cout<<endl;
        }
        // --------------------------------------------
```

```
    // ---------------------------------------------
    int * ptr ;
    for (int i = 0; i < 3; i++)
    {
        for (int j = 0; j < 3; j++)
        {
            ptr = &students[i][j];
            cout<<"Address is: "<<ptr<<endl;
        }
        cout<<endl;
    }
```

```cpp
student s1,s2,s3;
// -------------------
s1.name = "Talha";
s1.marks = 100;
s1.roll = 900;
// ------------------
s2.name = "Hamza";
s2.marks = 400;
s2.roll = 8880;
// --------------------
s3.name = "Ali";
s3.marks = 7600;
s3.roll = 120;
cout<<"-------------------------"<<endl;
cout<<"Student 1 Name: "<<s1.name<<endl;
cout<<"Student 1 Marks: "<<s1.marks<<endl;
cout<<"Student 1 Roll:  "<<s1.roll<<endl;
cout<<"-------------------------"<<endl;
cout<<"-------------------------"<<endl;
cout<<"Student 2 Name: "<<s2.name<<endl;
cout<<"Student 2 Marks: "<<s2.marks<<endl;
cout<<"Student 2 Roll:  "<<s2.roll<<endl;
cout<<"-------------------------"<<endl;
cout<<"-------------------------"<<endl;
cout<<"Student 3 Name: "<<s3.name<<endl;
cout<<"Student 3 Marks: "<<s3.marks<<endl;
cout<<"Student 3 Roll:  "<<s3.roll<<endl;
cout<<"-------------------------"<<endl;
```

```cpp
int students[3][3] = {
    {80, 85, 78},
    {88, 82, 90},
    {75, 89, 84}};
int n = 3;
for (int u = 0; u < 3; u++)
{
    for (int j = 0; j < n - u - 1; j++)
    {
        if (students[u][j] > students[u][j + 1])
        {
            swap(students[u][j], students[u][j + 1]);
        }
    }
}
for (int u = 0; u < 3; u++)
{
    for (int j = 0; j < 3; j++)
    {
        cout << students[u][j] << " ";
    }
    cout << endl;
}
```

Task # 04

```cpp
#include <iostream>
using namespace std;

struct Node
{
    string names;
    Node *next;
};

void insertion_beg(Node *&head, string name)
{
    Node *newNode = new Node();
    newNode->names = name;

    if (head == NULL)
    {
        head = newNode;
        return;
    }
    newNode->next = head;
    head = newNode;
}
```

```
void insertion_End(Node *&head, string name)
{
    Node *newNode = new Node();
    newNode->names = name;
    newNode->next = NULL;

    if (head == NULL)
    {
        insertion_beg(head, name);
        return;
    }
    Node *temp = head;
    while (temp->next != NULL)
    {
        temp = temp->next;
    }
    temp->next = newNode;
}
```

```
void insertion_pos(Node *&head, string name, int pos)
{
    Node *newNode = new Node();
    newNode->names = name;
    if (pos == 1)
    {
        insertion_beg(head, name);
        return;
    }
    Node *temp = head;
    for (int i = 0; i < pos - 1; i++)
    {
        temp = temp->next;
    }
    newNode->next = temp->next;
    temp->next = newNode;
}
```

```cpp
void display(Node *head)
{
    if (head == NULL)
    {
        cout << "List is Empty." << endl;
        return;
    }
    Node *temp = head;
    while (temp != NULL)
    {
        cout << temp->names << " -> ";
        temp = temp->next;
    }
    cout<<"NULL";
}
```

```cpp
int main()
{
    // ---------------------------Task # 04 ---------------------------
    Node *head = NULL;
    int choice = 0;
    string name;

    while (choice != 5)
    {
        cout << "\n1: Insertion at beg: " << endl;
        cout << "2: Insertion at End: " << endl;
        cout << "3: Insertion at Pos: " << endl;
        cout << "4: Display: " << endl;
        cout << "5: Exit: " << endl;
        cout << "Enter your Choice: ";
        cin >> choice;
```

```
while (choice != 5)
    cin >> choice;

    switch (choice)
    {
    case 1:
        cout << "Enter Name: ";
        cin >> name;
        insertion_beg(head, name);
        break;
    case 2:
        cout << "Enter Name: ";
        cin >> name;
        insertion_End(head, name);
        break;
    case 3:
        int pos;
        cout << "Enter Name: ";
        cin >> name;
        cout << "Enter Position: ";
        cin >> pos;
        insertion_pos(head, name, pos);
        break;
```

```
    switch (choice)
            int pos;
            cout << "Enter Name: ";
            cin >> name;
            cout << "Enter Position: ";
            cin >> pos;
            insertion_pos(head, name, pos);
            break;
        case 4:
            display(head);
            break;
        case 5:
            cout << "Exiting..." << endl;
            break;
        default:
            cout << "Wrong Choice." << endl;
            break;
        }
    }
```

```
1: Insertion at beg:
2: Insertion at End:
3: Insertion at Pos:
4: Display:
5: Exit:
Enter your Choice: 1
Enter Name: Talha

1: Insertion at beg:
2: Insertion at End:
3: Insertion at Pos:
4: Display:
5: Exit:
Enter your Choice: 2
Enter Name: Rafay

1: Insertion at beg:
2: Insertion at End:
3: Insertion at Pos:
4: Display:
5: Exit:
Enter your Choice: 3
Enter Name: Danish
Enter Position: 2
```

```
Enter your Choice: 4
Talha -> Rafay -> Danish -> NULL
```

Output of Task 1

```
Enter Quantity: 32

Enter Price: 21
Total : 672
```

```
Sum: 99
Sub: -79
Multiply: 890
Division: 0
Modulo: 10
```

```
Enter marks: 34
Your grade is: F
```

```
Number: 1
Number: 2
Number: 3
Number: 4
Number: 5
Number: 6
Number: 7
Number: 8
Number: 9
Number: 10
```

```
Enter Number: 43
Sum is: 946
```

```
Enter Number: 3
3 * 1 = 3
3 * 2 = 6
3 * 3 = 9
3 * 4 = 12
3 * 5 = 15
3 * 6 = 18
3 * 7 = 21
3 * 8 = 24
3 * 9 = 27
3 * 10 = 30
```

Output of Task # 02

```
Enter Number to Search:
15
NuMber is founded.
```

```
Enter Position:
3
5 15 22 66 35 40 55 65 70 85 95
```

```
Enter Position:
4
15 22 35 40 40 55 65 70 85
```

```
Total size of operator is: 40
```

```
Max: 95
Min: 5
```

Output of Task # 03

```
80 85 78
88 82 90
75 89 84
```

```
Address is: 0x1bcbff90c
Address is: 0x1bcbff910
Address is: 0x1bcbff914

Address is: 0x1bcbff918
Address is: 0x1bcbff91c
Address is: 0x1bcbff920
```